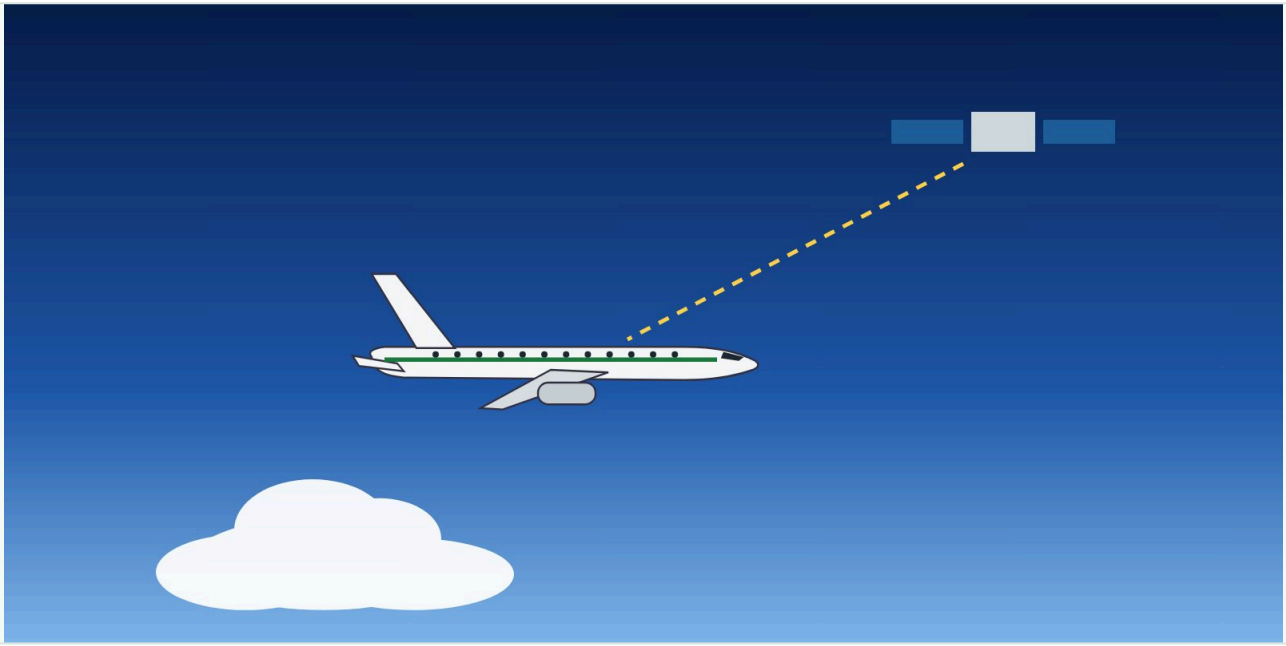
MODULE 1 · PBN PRINCIPLES

Lesson 02: GNSS, Integrity and the Navigation Database

Lesson 2 of 5 · about 7 min

LEARNING OBJECTIVES

- Explain GNSS integrity monitoring and its limits.
- Recognise GNSS interference threats.
- Check and use the navigation database correctly.



GNSS integrity — and interference — must be monitored.

GNSS and integrity

- Most PBN operations rely on **GNSS**. Integrity is provided by **receiver autonomous integrity monitoring (RAIM)** or equivalent aircraft-based augmentation, and by satellite-based augmentation (SBAS) where available.
- Check the **GNSS NOTAMs** and the predicted RAIM availability for the route and the time of the approach, as the company procedure requires.
- Know the backup: DME/DME or IRS updating, and conventional procedures and navaids.

GNSS interference

Jamming (signal blocked) and **spoofing** (false signal) are increasingly common in some regions. Effects include loss of GNSS position, position or time jumps, EGPWS terrain alerts, ADS-B and clock errors. Follow the company and manufacturer guidance: cross-check position with other sources, consider deselecting GNSS where the procedure allows, use conventional navaids, inform ATC and report.

The navigation database

- Check the **AIRAC cycle** is current before flight.
- Procedures must be **loaded from the database** by name — never built manually from waypoints for an RNP approach.
- Cross-check the loaded procedure with the chart: waypoint sequence, tracks, distances, altitude and speed constraints, and the vertical path angle on the final approach.
- Do not modify the final approach segment of an RNP approach (except where your procedure allows specific changes).

KEY TAKEAWAYS

- Check GNSS NOTAMs and RAIM / integrity availability.
- Jamming and spoofing: cross-check, use conventional navaids, inform ATC and report.
- Current AIRAC; load procedures by name; cross-check with the chart; do not modify the final approach.

KNOWLEDGE CHECK

How must an RNP approach be loaded in the FMS?

A. Built from individual waypoints

B. Loaded by name from the current navigation database and cross-checked with the chart

C. Copied from the previous flight

D. Entered as a direct-to

Answer: B. RNP approaches must be loaded from the database and checked against the chart.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.